RAVI NEWS

Proof of Concept Application

Technical Documentation & Build Report

Prepared for Client Review

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Table of Contents

1. Executive Summary
2. Project Overview
3. Target Audience & Design Philosophy
4. Technology Stack
5. Application Architecture
6. Reader App — Screens
7. Admin Panel — Screens
8. Key Features
9. Data Flow & Sync
10. File Structure & How to Run
11. Future Roadmap

1. Executive Summary

Ravi News is a hyperlocal news application designed specifically for Tier 2 and Tier 3 urban villages in the Nizamabad and Kamareddy districts of Telangana. The app delivers news in both Telugu and English, with an Inshorts-style swipeable card interface optimized for users who may have limited smartphone experience.

This document presents the Proof of Concept (POC) build, covering the **Reader App** for end users and the **Admin Panel** for news reporters. The POC demonstrates the core user experience, bilingual content delivery, and the complete news publishing workflow from reporter to reader.

2. Project Overview

Project Name	Ravi News
Type	Hyperlocal News Application (Mobile-First Web App)
Region	Nizamabad & Kamareddy Districts, Telangana
Languages	Telugu and English (bilingual toggle)
Inspiration	Inshorts, Way2News, Mee24News
POC Status	Fully functional interactive prototype

The app fills a gap in hyperlocal news coverage for mandal-level communities. Existing news apps focus on state or national news, while Ravi News delivers ward-level, mandal-level, and district-level stories that directly impact local residents.

3. Target Audience & Design Philosophy

3.1 Target Users

- Residents of Tier 2 and Tier 3 urban villages in Telangana
- Primarily Telugu-speaking population with varying English literacy
- Users with basic smartphones and moderate internet connectivity
- Age group: 18-65, emphasis on ease of use for all age groups

3.2 Design Principles

1. **Super Minimal:** Inspired by Inshorts and Way2News. One news story per screen, swipe to navigate.

2. **Telugu First:** App defaults to Telugu with easy toggle to English.

3. **Mobile-First:** Designed for 390px phone screens. Touch swipe + keyboard navigation.

4. **Fast & Lightweight:** No heavy frameworks. Pure React with zero external UI dependencies.

4. Technology Stack

Component	Technology
Frontend Framework	React 19.2 (Functional Components + Hooks)
Build Tool	Vite 8.0 (Beta)
Language	JavaScript (JSX)
Styling	Inline CSS (zero external dependencies)
Data Store	localStorage (shared cross-tab)
Routing	Hash-based routing (#admin)
Sync Mechanism	Custom events + storage events + polling
Image Assets	Unsplash stock photos (placeholder)

5. Application Architecture

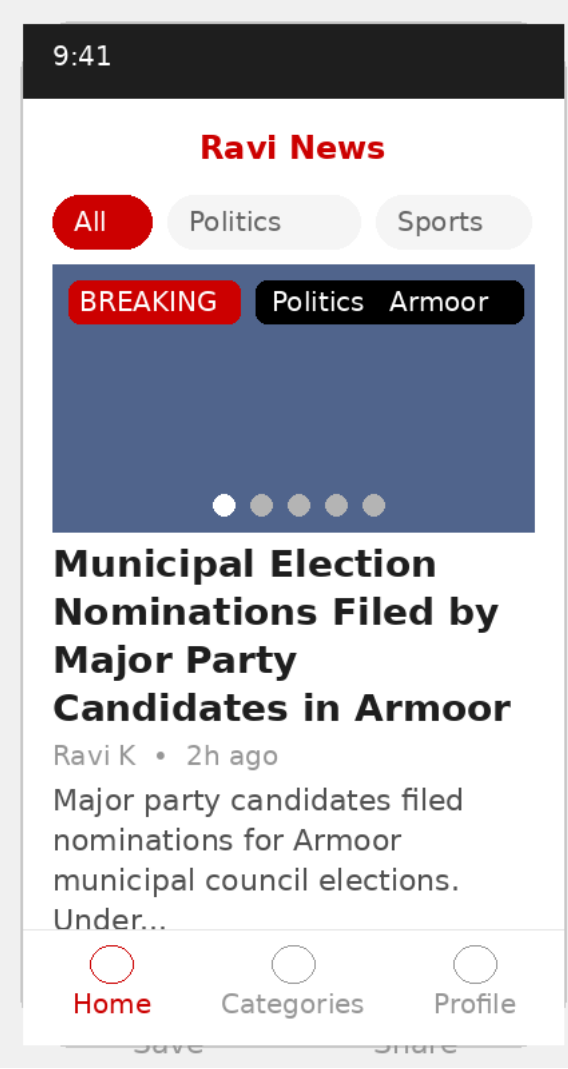
The Ravi News POC is a single-page application (SPA) with two main views served from the same codebase, separated by hash-based routing:

URL	View	Purpose
`localhost:5173/`	Reader App	End-user news consumption
`localhost:5173/#admin`	Admin Panel	Reporter/editor content management

Both views share a common data store (**store.js**) backed by localStorage. When a reporter publishes an article in the Admin Panel, it becomes immediately visible in the Reader App through a real-time sync mechanism combining custom DOM events, storage events, and 3-second polling.

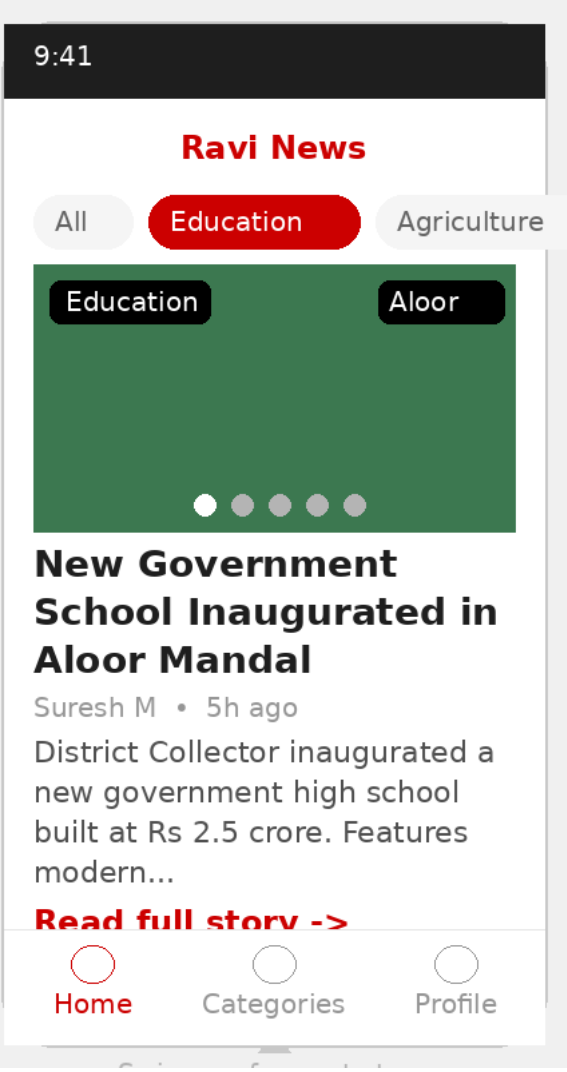
6. Reader App — Screens

The reader app follows an Inshorts-style full-screen card design. Below are all the key screens with their phone mockups showing exactly how the app appears on a mobile device.



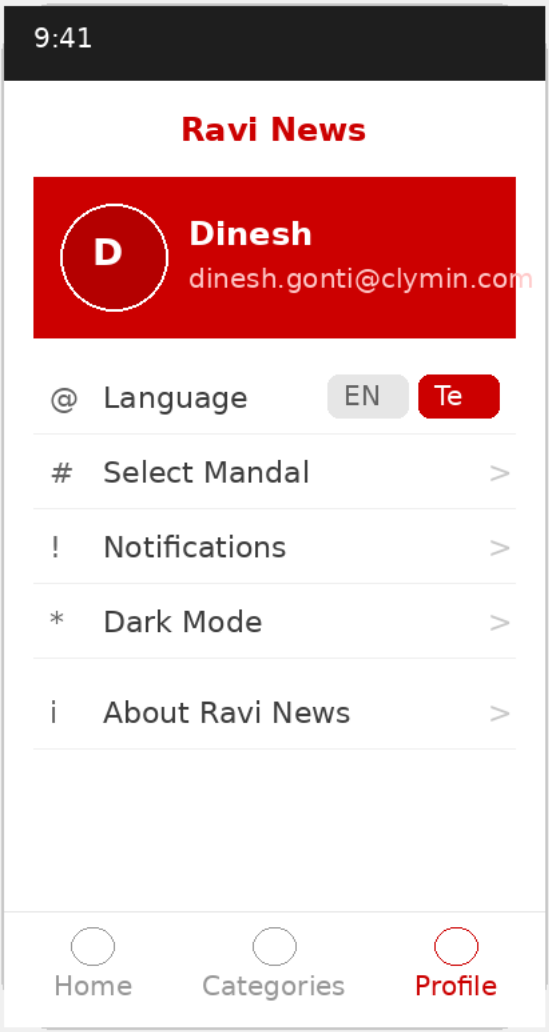
6.1 Home Feed — Breaking News

Full-screen swipeable news card with BREAKING badge, category pill, mandal location tag, and swipe hint.



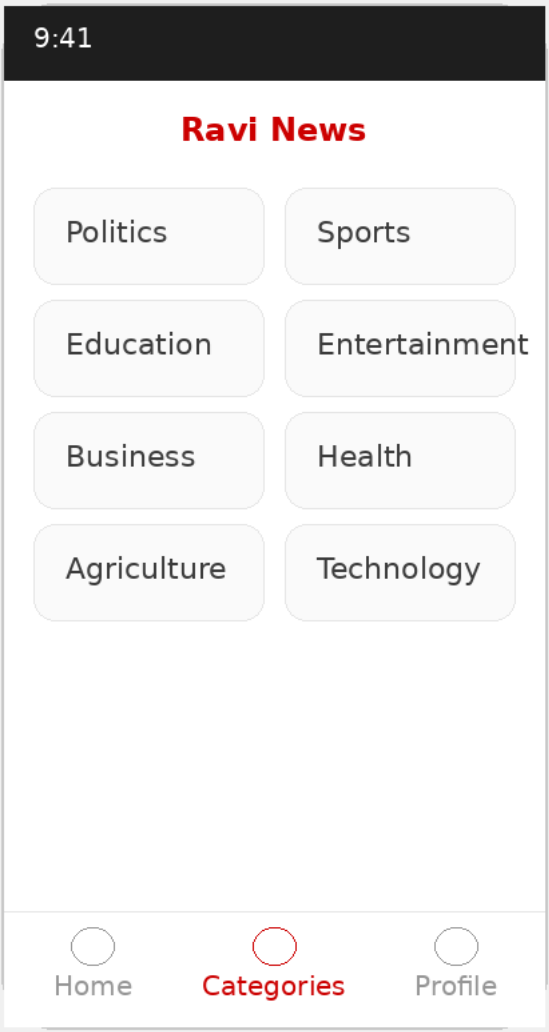
6.2 Home Feed — Education Category

Category-filtered view showing Education news. Same card layout with different content and category highlight.



6.3 Profile & Settings

User profile with EN/Telugu toggle, mandal selection, notifications, dark mode, and app info.

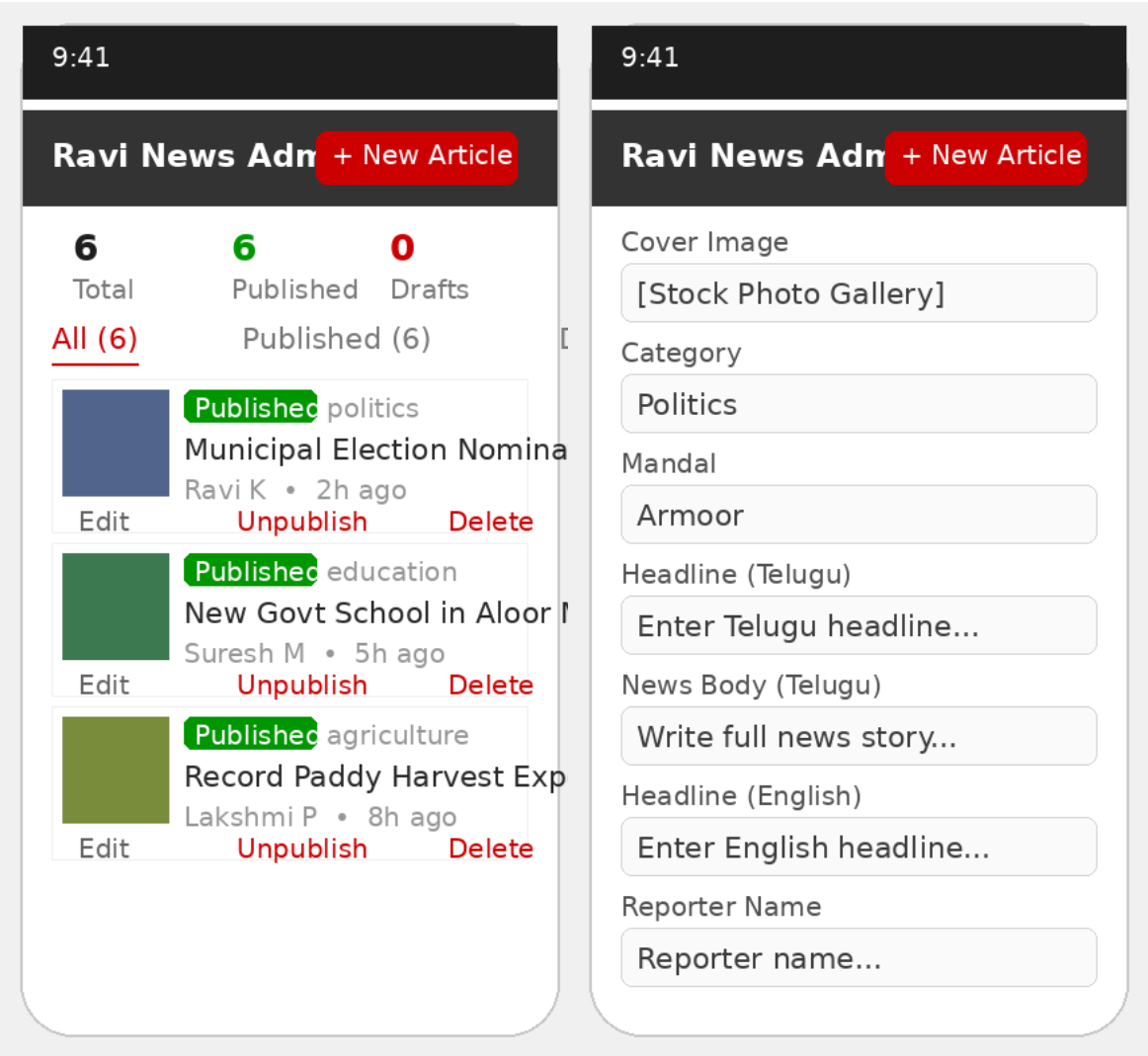


6.4 Categories Grid

2-column grid of all news categories. Tap any category to filter the home feed.

7. Admin Panel — Screens

The admin panel is a web-based CMS for reporters and editors to manage news articles. It provides full CRUD operations with real-time sync to the reader app.



7.1 Admin Dashboard

Article list with stats, filter tabs, status badges, and CRUD actions (Edit, Unpublish, Delete).

7.2 New / Edit Article Form

Article creation form with image picker, category/mandal dropdowns, bilingual content tabs, and live preview.

8. Key Features

Feature	Description
Bilingual Support	Full Telugu + English. Instant toggle switches all UI, categories, and content.
Inshorts-Style Cards	One story per screen. Swipe/drag/arrow navigation. Full-screen immersive reading.
Breaking News	Visual indicator with lightning bolt icon and red badge on urgent stories.
Category Filtering	Scrollable pills + dedicated grid. Filter news by Politics, Sports, Education, etc.
Mandal Locations	News tagged by mandal (Armoor, Aloor, Bodhan, etc.). Location badge on each card.
Admin CRUD	Full Create, Read, Update, Delete for articles. Publish/Draft toggle.
Real-Time Sync	Articles published in admin appear in reader within 3 seconds. No refresh needed.
Live Preview	Admin shows live card preview while editing articles.
Mobile-First	390px phone frame with touch gestures. Works on any mobile browser.

9. Data Flow & Sync

9.1 Data Store (store.js)

A centralized module managing all article data via localStorage. Exposes CRUD functions: **getArticles()**, **addArticle()**, **updateArticle()**, **deleteArticle()**. Default sample articles are seeded on first load covering 8 mandals and 6 categories.

9.2 Cross-Tab Synchronization

1. **Custom Events:** A 'ravi-news-updated' event fires on data save for same-tab listeners.
2. **Storage Events:** Browser's native storage event notifies other tabs when localStorage changes.
3. **Polling Fallback:** A 3-second interval catches any missed updates as a safety net.

10. File Structure & How to Run

10.1 File Structure

File	Purpose
`src/main.jsx`	Entry point with hash-based Router component
`src/App.jsx`	Reader app — Inshorts-style swipeable news cards
`src/Admin.jsx`	Admin panel — CRUD interface for reporters
`src/store.js`	Shared data store with localStorage persistence
`src/index.css`	Minimal CSS reset (all styles are inline)
`package.json`	Project config with React 19 + Vite 8

10.2 How to Run

1. Clone or download the **ravi-news-poc** project folder
2. Open terminal and navigate: `cd ravi-news-poc`
3. Install dependencies: `npm install`
4. Start development server: `npm run dev`
5. Open Reader App: `http://localhost:5173/`
6. Open Admin Panel: `http://localhost:5173/#admin`

11. Future Roadmap

Phase 2 — Backend & Database

• Cloud database (Firebase / Supabase / PostgreSQL)

• RESTful API for article management

• User authentication for reporters

Phase 3 — Native Mobile App

• React Native for Android and iOS

- Push notifications for breaking news
- Offline mode with cached articles

Phase 4 — Advanced Features

- Voice-to-text news reading in Telugu
- Video news clips integration
- Community comments and reactions
- Advertising platform for local businesses
- Analytics dashboard for article performance

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